

■ Project Assignment Report: Traffic Signal Synchronization System

Project Title: Traffic Signal Synchronization System Using Chinese Remainder Theorem

Course / Assignment: Number Theory & Discrete Mathematics Application

Team Leadership: Justin Joe Mathew (Team Lead)

■ Does Traffic Signal Synchronization Demo CRT or just LCM/GCD?

Mathematical Distinction:

1. LCM (Least Common Multiple) / GCD:

Used when all signals start at Minute 0 with NO remainder/offset. If Signal 1 (3 min), Signal 2 (5 min), and Signal 3 (7 min) all turn green at Minute 0, the next simultaneous green light is simply $\text{LCM}(3, 5, 7) = 105$ minutes.

2. CRT (Chinese Remainder Theorem):

REQUIRED when signals have DIFFERENT REMAINDERS / OFFSETS.

In our project scenario:

- Signal 1 (3-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod{3}$
- Signal 2 (5-min cycle) turned green 3 minutes ago $\rightarrow x \equiv 3 \pmod{5}$
- Signal 3 (7-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod{7}$

Because the remainders (2, 3, 2) are non-zero and distinct, LCM cannot solve this problem. The system requires the Chinese Remainder Theorem (Extended Euclidean Algorithm & Modular Inverses) to solve for the unique green wave minute ($x = 23$).

■ Conclusion: The project is a genuine demonstration of CRT, specifically showcasing why CRT is needed over simple LCM when real-world cycles start at different offsets.

■ Physical Hardware Prototype & Digital Twin Interface

The project includes both a physical Arduino Nano breadboard circuit and a full digital twin web simulator.

■ Interactive Web App & Project Repository: ``https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem`` (Local Dev Server: ``http://localhost:5173/``)

REAL-TIME SYNCHRONIZATION ENGINE

CRT Traffic Light Lab

Hardware digital twin mirroring an ATmega328P prototype. Demonstrates multi-frequency phase drift and Chinese Remainder Theorem convergence.

Audio Muted

Team (10)

Pinout Specs

RUNNING (DRIFTING)

Live Signal Array

ATmega328P View

SIG A3.0s

SIG B5.0s

SIG C7.0s

YELLOW

GREEN

GREEN

Pause

Force Re-Sync

Light Show

Reset

CRT Remainder & Period Combinations:

LCM = 105s

105s

30s

12s

CRT Remainder State Seeker

Solve for timestamp t

A: RED

B: RED

C: RED

Seek to Match

Match occurs at: t = 0.00s

CRT Telemetry Engine

105s Recurrence

PERIOD SET

3.0s / 5.0s / 7.0s

HYPERPERIOD LCM

105.00 s

CLOCK PROGRESS

34.55s / 105s

RECURRENCE INDEX

Cycle #3

CRT Modular Recurrence: $t \equiv 0 \pmod{LCM}$

$\Rightarrow LCM(3, 5, 7) = 105$ seconds

$\Rightarrow$ Cycles per LCM: A: 35 | B: 21 | C: 15

$\Rightarrow$ Buzzer chime: Only on synchronization ($t \equiv 0$)

Signal	Remainder ($t \bmod T$)	State
Sig A (3.0s)	1.55 s	YELLOW
Sig B (5.0s)	4.55 s	GREEN
Sig C (7.0s)	6.55 s	GREEN

Multi-Channel Phase Waveform

Window: 0.00s – 105.00s

While cursor sweeps across the 105s hyperperiod, Cyan vertical line marks simultaneous alignment.

SIG A

SIG B

SIG C

Arduino Nano Physical Firmware

Copy Arduino Code

arduino/traffic_light_controller.ino

Signal A: Pin 2 (Red), Pin 3 (Yellow), Pin 4 (Green) • Signal B: Pin 8 (Red), Pin 9 (Yellow), Pin 18 (Green) •

Signal C: Pin A5 (Red), Pin A4 (Yellow), Pin A3 (Green) • Buzzer: Pin A1

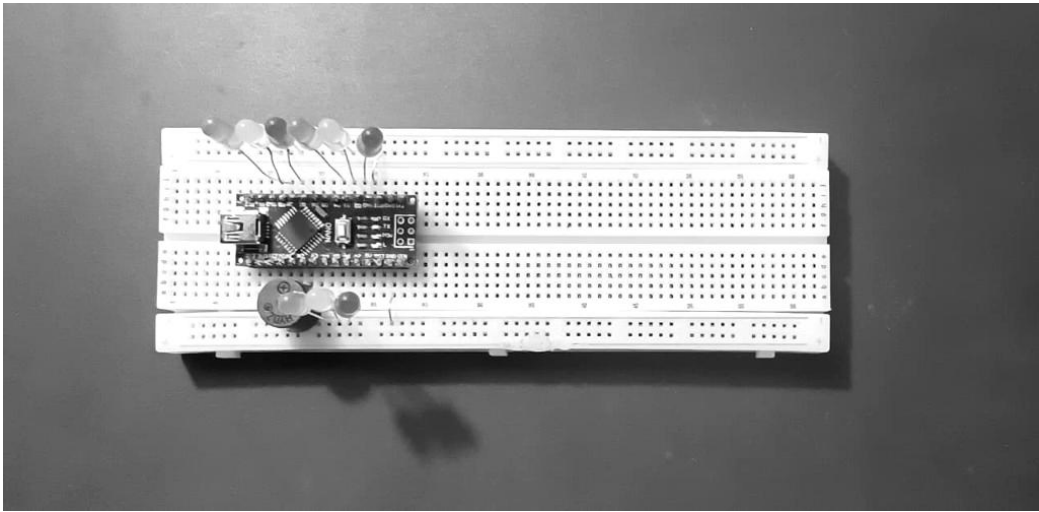
```
const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 18;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

const unsigned long PERIOD_A = 3000;
const unsigned long PERIOD_B = 5000;
const unsigned long PERIOD_C = 7000;
const unsigned long MASTER_CYCLE = 105000;
```



■ Group Member Roles & Work Division (10 Members)

PROJECT TEAM STRUCTURE		
■ 1. Project Leadership & Model	■ Justin Joe Mathew (Team Lead) & Jyothis Liju	■
■ 2. Hardware Sourcing & Build Testing	■ Joseph Alex & Joseph J	■
■ 3. Web Design & Frontend	■ Johan Geo & Johaan Sam	■
■ 4. Presentation & Showcase	■ Joel Geo Manuel	■
■ 5. Mathematics & Applications	■ Joel Duke, Jyothika Prakash, & Jose Alex	■

■ Detailed Team Member Contribution Breakdown

1. Project Leadership & Working Model Construction

■ Team Members: Justin Joe Mathew (Team Lead) & Jyothis Liju

- Justin Joe Mathew (Team Lead): Guided overall project direction, engineered core system architecture, and led the construction and logic of the working traffic signal model.
- Jyothis Liju: Assisted Justin directly in constructing the working traffic signal model, testing signal synchronization alignment, and verifying physical component interaction.

2. Hardware Sourcing, Build Testing & Application Design

■ Team Members: Joseph Alex & Joseph J

- Joseph Alex: Determined component requirements, purchased hardware parts, executed physical build testing, and contributed to application use-cases.
- Joseph J: Handled hardware part selection, procurement, mechanism assembly testing, and co-developed application integration ideas.

3. Web Design & Frontend Development

■ Team Members: Johan Geo & Johaan Sam

- Johan Geo: Designed the user interface, visual layout, traffic signal dashboard theme, and interactive monitor dials.
- Johaan Sam: Built the frontend components, state integration, responsive layouts, and interactive controls.

4. Presentation & Demo Lead

■ Team Member: Joel Geo Manuel

- Joel Geo Manuel: Handled the presentation structure, demo walkthrough flow, visual slides, and showcase delivery for explaining the project to an audience.

5. Mathematics & CRT Applications Research

■ Team Members: Joel Duke, Jyothika Prakash, & Jose Alex

- Joel Duke: Researched linear congruences and formulated the traffic signal synchronization modulo equations ($3, 5, 7$ minute cycles).

- Jyothika Prakash: Explored real-world CRT applications and authored step-by-step mathematical explanations and hint structures.

- Jose Alex: Investigated Chinese Remainder Theorem proofs, solution bounds ($N = 105$), and modulo remainder properties.